

Fraction-of-50 Structure in CPTG and Related Cosmological Diagnostics

Research note with source citations

Prepared as a diagnostic catalogue for Curvature Polarization Transport Gravity (CPTG)

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Abstract. This note catalogues denominator-50 and nearby compact-rational structures in the CPTG galaxy-limit equations, the reduced transport framework, an exploratory early-expansion CMB transport-source run, and selected cosmological quantities not specific to CPTG. Equation-level CPTG constants are separated from dimensionless diagnostics and from rounded statistical outputs. External cosmological entries are included as comparative references only; they should not be read as evidence for CPTG unless they survive model-specific ablation, null testing, and independent reruns.

1 Scope and interpretation rules

A denominator-50 lattice is defined by

$$x_n = \frac{n}{50}, \quad n \in \mathbb{Z}. \quad (1)$$

The catalogue records quantities that lie directly on this lattice, can be rewritten on it, or recur near nearby compact rational values. The evidentiary hierarchy is:

$$\text{equation-level constants} > \text{dimensionless physical ratios} > \text{rounded fit statistics}. \quad (2)$$

The CPTG-specific equations are drawn from the uploaded CPTG manuscript, especially the reduced suppression law, galaxy-limit equation, baryonic source mapping, geometric transport remnant, structural mode definition, outer-regime attractor, and BTFR derivation [1]. CMB-source entries are drawn from the uploaded early-transport source-model summary [2]. External cosmological entries use Planck 2018 results, MOND literature, BTFR literature, and inflationary e-fold references [3, 4, 5, 6, 7, 8].

2 Equation-level fraction structure in the CPTG paper

2.1 Reduced suppression law

The CPTG transport suppression factor is written as [1]

$$\Sigma(g) = \left[1 + \left(\frac{g}{a_\star} \right)^{123/50} \right]^{-11/50}. \quad (3)$$

The reduced galaxy-limit equation carries the same fractional suppression structure:

$$g(r) = g_{\text{bar}}(r) + \frac{\sqrt{a_\star g_{\text{bar}}(r)} + g_{\text{geom}}(r)}{\left[1 + (g(r)/a_\star)^{123/50} \right]^{11/50}}. \quad (4)$$

The exponents

$$\frac{123}{50} = 2.46, \quad \frac{11}{50} = 0.22 \quad (5)$$

are therefore equation-level response constants rather than visual pattern matches or output-format artifacts.

2.2 CPTG-weighted baryonic source field

In the reduced galaxy-limit source mapping, the gas, disk, and bulge components are weighted as [1]

$$g_{\text{bar}}(r) = \frac{\frac{27}{50}v_{\text{gas}}^2(r) + \frac{23}{50}v_{\text{disk}}^2(r) + \frac{32}{50}v_{\text{bul}}^2(r)}{r}. \quad (6)$$

The gas and disk coefficients form a closed complementary pair:

$$\frac{27}{50} + \frac{23}{50} = 1, \quad \frac{1}{2} \left(\frac{27}{50} + \frac{23}{50} \right) = \frac{25}{50} = \frac{1}{2}. \quad (7)$$

The bulge coefficient is a compact-source weighting equal to 16/25, expressed above on the denominator-50 lattice as 32/50. It should be treated separately from the gas-disk closure pair.

2.3 Geometric transport remnant

The radial geometric transport remnant is [1]

$$g_{\text{geom}}(r) = \eta a_{\star} \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}, \quad \eta = \frac{1}{2}. \quad (8)$$

The powers and the averaging coefficient can be rewritten as

$$\frac{7}{5} = \frac{70}{50}, \quad \frac{12}{5} = \frac{120}{50}, \quad \eta = \frac{25}{50}. \quad (9)$$

These are equation-level radial-shape quantities. They are stronger evidence than post-fit numerical coincidences because they participate directly in the reduced field response.

2.4 CPTG scale relation and circular normalization

The CPTG acceleration scale is linked to the cosmological scale by [1]

$$a_{\star} = \frac{3}{16\pi} c H_0, \quad H_0 = \frac{16\pi a_{\star}}{3 c}. \quad (10)$$

The coefficient

$$\frac{3}{16\pi} \simeq 0.05968 \simeq \frac{3}{50} \quad (11)$$

is not a denominator-50 constant in exact form, but it falls very close to a compact denominator-50 value because of the inverse circular normalization. This is best interpreted as a circular-normalization connection, not as an additional exact CPTG lattice coefficient.

3 Outer regime, structural mode, and BTFR scaling

3.1 Outer-regime attractor

The extended CPTG solution defines the local logarithmic rotation-curve slope as [1]

$$s(r) \equiv \frac{d \ln V}{d \ln r}, \quad s_{\infty} = \frac{1}{4}. \quad (12)$$

Equivalently,

$$V(r) \propto r^{1/4}, \quad g(r) \propto r^{-1/2}. \quad (13)$$

This is a quarter-power asymptotic attractor. It is not an integer $n/50$ entry, but it is part of the same compact-fraction response family and arises from the square-root low-acceleration polarization scaling.

3.2 Structural mode interval

The Curvature-Weighted Structural Mode Index is [1]

$$N \equiv \frac{R}{\lambda}. \quad (14)$$

With

$$w(r) = \left| \frac{dC}{dr} \right|, \quad (15)$$

the outer curvature-organization scale is

$$\lambda = \frac{\int_{R/5}^R r w(r) dr}{\int_{R/5}^R w(r) dr}, \quad N = \frac{R}{\lambda}. \quad (16)$$

The lower radial cutoff $R/5$ can be written as $10/50$ of the resolved radius. This is a weak but explicit denominator-50-compatible interval in the structural-mode definition.

3.3 BTFR scaling

The baryonic Tully-Fisher relation is empirically expressed as a fourth-power relation between baryonic mass and characteristic rotation speed [7]. In CPTG, the leading low-acceleration scaling is [1]

$$g(r) \simeq \sqrt{a_\star g_{\text{bar}}(r)}. \quad (17)$$

Using

$$g_{\text{bar}}(r) \simeq \frac{GM_{\text{eff}}}{r^2}, \quad g(r) = \frac{V^2(r)}{r}, \quad (18)$$

the radial dependence cancels and yields

$$V_f^4 \simeq Ga_\star M_{\text{eff}}. \quad (19)$$

With $M_{\text{eff}} = \Xi M_b$, this becomes

$$V_f^4 \simeq Ga_\star \Xi M_b. \quad (20)$$

The fourth-power BTFR slope is therefore tied to the square-root polarization response and the CPTG source mapping, rather than imposed as a separate interpolation law.

4 CPTG CMB-source model diagnostics

4.1 Early-expansion clock values

The exploratory CMB transport-source summary records a Planck-like early-expansion clock with [2]

$$H_0 = 67.321170, \quad \Omega_m = 0.31582310, \quad \Omega_r = 9.27233359 \times 10^{-5}, \quad (21)$$

$$z_\star = 1089.914, \quad z_{\text{eq}} = 3405.080. \quad (22)$$

The equality-to-recombination ratios are

$$\frac{z_\star}{z_{\text{eq}}} = \frac{1089.914}{3405.080} \simeq 0.32008, \quad (23)$$

$$\frac{1+z_\star}{1+z_{\text{eq}}} \simeq 0.32028, \quad \frac{1+z_{\text{eq}}}{1+z_\star} \simeq 3.12223. \quad (24)$$

The inverse interval is close to π at the percent level. This makes the equality-to-recombination timing interval a useful diagnostic of compact inverse-circular scaling, independent of any claim that it is a CPTG constant.

4.2 CMB shell-geometry ratios

The best raw hard-shell source proxy and best cross-validated proxy give compact shell geometries [2]:

$$r_{\text{shell,raw}} = \frac{178.370773}{552.181947} \simeq 0.3230, \quad (25)$$

$$r_{\text{shell,CV}} = \frac{176.977272}{539.719208} \simeq 0.3279. \quad (26)$$

These ratios compare two fitted shell-scale quantities and therefore form dimensionless geometry. They are stronger diagnostics than rounded χ^2 values, but still require seed-repeat, block-holdout, and full-pipeline null tests.

4.3 CMB fit and validation scores

The same run reports positive TT, TE, and EE holdout improvements, with the best source proxy exceeding the calibrated 95% look-elsewhere null but not the 99% level [2]. The score values align closely with a denominator-50 reporting grid:

$$\Delta\chi_{\text{TT}}^2 \simeq 7.7774 \simeq \frac{389}{50}, \quad \Delta\chi_{\text{TE}}^2 \simeq 7.3556 \simeq \frac{368}{50}, \quad (27)$$

$$\Delta\chi_{\text{EE}}^2 \simeq 2.0048 \simeq \frac{100}{50}, \quad \sum \Delta\chi^2 \simeq 17.1377 \simeq \frac{857}{50}. \quad (28)$$

These entries are retained as weak diagnostics only. Statistical scores can land near simple grids through rounding and should not be weighted like equation-level constants or dimensionless shell geometry.

5 External cosmological connections not specific to CPTG

5.1 Planck 2018 parameter values

Planck 2018 reports, for base Λ CDM, representative values including $\Omega_c h^2 \simeq 0.120$, $\Omega_b h^2 \simeq 0.0224$, $n_s \simeq 0.965$, optical depth $\tau \simeq 0.054$, $100\theta_* \simeq 1.0411$, $H_0 \simeq 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m \simeq 0.315$, $\sigma_8 \simeq 0.811$, and $N_{\text{eff}} \simeq 2.99$ in the relevant combined analyses [3]. Several of these lie near denominator-50 grid values:

$$\Omega_c h^2 \simeq 0.120 \simeq \frac{6}{50}, \quad \Omega_b h^2 + \Omega_c h^2 \simeq 0.1425 \simeq \frac{7}{50}, \quad (29)$$

$$\Omega_m \simeq 0.315 \simeq 0.32, \quad \Omega_\Lambda \simeq 0.685 \simeq \frac{34}{50}, \quad (30)$$

$$\frac{\Omega_b h^2}{\Omega_b h^2 + \Omega_c h^2} \simeq 0.157 \simeq \frac{8}{50}, \quad \frac{\Omega_c h^2}{\Omega_b h^2 + \Omega_c h^2} \simeq 0.843 \simeq \frac{42}{50}. \quad (31)$$

Additional weaker grid-near entries include

$$n_s \simeq 0.965 \simeq \frac{48}{50}, \quad 100\theta_* \simeq 1.0411 \simeq \frac{52}{50}, \quad (32)$$

$$\sigma_8 \simeq 0.811 \simeq \frac{41}{50}, \quad \tau \simeq 0.054 \simeq \frac{3}{50}, \quad N_{\text{eff}} \simeq 2.99 \simeq \frac{150}{50}. \quad (33)$$

These external entries should be treated cautiously. Some are physical density parameters; others depend on conventional scaling choices, such as the factor of 100 in $100\theta_*$.

5.2 Matter-radiation composition at recombination

Using the equality-to-recombination scale-factor ratio in the CMB proxy, the matter-to-radiation ratio at recombination is

$$\left. \frac{\rho_m}{\rho_r} \right|_{z_*} = \frac{1 + z_{\text{eq}}}{1 + z_*} \simeq 3.12223 \simeq \frac{156}{50} \simeq \pi. \quad (34)$$

The corresponding matter and radiation fractions within the matter-plus-radiation budget are

$$f_m \simeq 0.7574 \simeq \frac{38}{50}, \quad f_r \simeq 0.2426 \simeq \frac{12}{50}. \quad (35)$$

This is an external cosmological diagnostic derived from equality and recombination timing. It is not a CPTG equation-level constant, but it is relevant because CPTG’s exploratory CMB program is explicitly focused on early-expansion transport before recombination.

5.3 MOND acceleration scale

MOND introduces a characteristic acceleration scale, conventionally quoted near [5, 6]

$$a_0 \simeq 1.2 \times 10^{-10} \text{ m s}^{-2}. \quad (36)$$

The dimensionless coefficient is

$$1.2 = \frac{60}{50}. \quad (37)$$

This does not imply that MOND is a denominator-50 theory. It simply places the standard MOND acceleration coefficient on the same numerical reporting lattice used in this diagnostic catalogue.

5.4 Inflationary e-fold context

Inflationary cosmology often discusses observable perturbations in terms of roughly 50 to 60 e-folds before the end of inflation, with the exact number depending on assumptions about reheating and post-inflationary expansion [8, 4]. This is not a denominator-50 fraction, but it is a direct recurrence of the number 50 in early-universe scaling arguments.

$$N_{\text{obs}} \sim 50\text{--}60. \quad (38)$$

This entry is included for completeness only and should be classified as contextual, not as evidence for CPTG fraction structure.

6 Pi and circular normalization

The denominator-50 lattice approximates common circular constants at two-decimal precision:

$$\pi \simeq \frac{157}{50} = 3.14, \quad 2\pi \simeq \frac{314}{50} = 6.28. \quad (39)$$

The inverse circular scale

$$\frac{1}{2\pi} \simeq 0.15915 \simeq \frac{8}{50} \quad (40)$$

is especially useful as a comparison point because baryon fractions and circular-normalization coefficients can land near it. The CPTG scale relation also contains $3/(16\pi)$, which falls near $3/50$.

7 Diagnostic catalogue

Table 1: CPTG and cosmology-side fraction diagnostics. Strength labels distinguish equation-level constants from dimensionless ratios and rounded outputs.

Quantity	Value	Grid form	Status and source
Suppression exponent	123/50	123/50	Equation-level CPTG response exponent [1].
Suppression power	11/50	11/50	Equation-level CPTG response exponent [1].
Gas source weight	27/50	27/50	Equation-level baryonic source coefficient [1].
Disk source weight	23/50	23/50	Equation-level baryonic source coefficient [1].
Bulge source weight	32/50	32/50	Equation-level compact-source coefficient [1].

Quantity	Value	Grid form	Status and source
Gas-disk closure	27/50 + 23/50	50/50	Complementary pair; high diagnostic value.
Gas-disk center	(27/50 + 23/50)/2	25/50	Central local-source response scale.
Transport exponent	7/5	70/50	Equation-level geometric remnant power [1].
Transport denominator exponent	12/5	120/50	Equation-level geometric remnant power [1].
Transport coefficient	1/2	25/50	Equation-level transport remnant coefficient [1].
Scale coefficient	3/(16 π)	near 3/50	Circular-normalized CPTG scale coefficient [1].
Structural-mode cutoff	$R/5$	10/50 of R	Explicit interval in structural-mode definition [1].
Outer attractor	1/4	quarter-power	Compact asymptotic slope; not integer $n/50$ [1].
CDM physical density	0.120	6/50	External Planck-like parameter [3].
Matter physical density	0.1425	near 7/50	External Planck-like derived value [3].
Matter density fraction	0.315–0.316	near 0.32	External cosmological ratio [3].
Dark-energy fraction	0.684–0.685	near 34/50	External complementary ratio [3].
Baryon matter fraction	0.157	near 8/50	External matter-composition ratio [3].
CDM matter fraction	0.843	near 42/50	External matter-composition ratio [3].
Recombination/equality ratio	0.3203	near 0.32	CMB timing ratio from proxy values [2].
Equality/recombination expansion	3.1222	near 156/50	Close to π ; external timing diagnostic.
Matter fraction at recombination	0.7574	near 38/50	Derived composition at recombination.
Radiation fraction at recombination	0.2426	near 12/50	Complementary derived composition.
Raw CMB shell ratio	0.3230	0.32-class	Dimensionless shell geometry [2].
CV CMB shell ratio	0.3279	0.32-class to 1/3	Cross-validated shell geometry [2].
Raw transport fraction	0.5532	near 28/50	Model diagnostic; medium strength [2].
CV transport fraction	0.5258	near 26/50	Model diagnostic; medium strength [2].
TT holdout score	7.7774	near 389/50	Weak; rounded statistical output [2].
TE holdout score	7.3556	near 368/50	Weak; rounded statistical output [2].
EE holdout score	2.0048	near 100/50	Weak; rounded statistical output [2].
Holdout sum	17.1377	near 857/50	Weak; rounded statistical output [2].
MOND coefficient	1.2	60/50	External modified-gravity scale coefficient [5, 6].
π	3.14159	near 157/50	Circular normalization benchmark.
2π	6.28318	near 314/50	Circular normalization benchmark.
$1/(2\pi)$	0.15915	near 8/50	Inverse circular normalization.
Inflation e-fold context	$N \sim 50\text{--}60$	direct 50-scale	Contextual early-universe recurrence [8, 4].

8 Ablation and validation program

The galaxy-limit constants should be protected during exploratory development. If a one-step perturbation of an equation-level denominator-50 source or response constant degrades the benchmark sharply, the appropriate research action is to document the sensitivity as an ablation result rather than to tune new constants toward the lattice.

Recommended galaxy-limit ablations:

1. Perturb each equation-level coefficient by one lattice step and report the benchmark change.
2. Test the gas-disk pair with closure preserved and closure broken.
3. Test the compact bulge coefficient independently from the gas-disk pair.
4. Recompute SPARC benchmark margins against MOND and MOND-plus- Υ for every ablation.
5. Repeat with random seeds, larger galaxy subsets, and independent distance-quality cuts.

Recommended CMB and cosmology-side tests:

1. Keep denominator-50 tracking diagnostic-only in all interim source-proxy models.
2. Report nearest-lattice distance for dimensionless structural ratios, not only for raw χ^2 scores.
3. Add block holdouts in low- ℓ , shell-region, and high- ℓ intervals.

4. Add full-pipeline null tests, not only sign-flip nulls on stored templates.
5. Compare free, clock-tied, and constrained shell models only in the final revision stage.

9 Research interpretation

The strongest denominator-50 structures are the constants embedded directly in the CPTG equations: the suppression exponents, the baryonic source weights, the radial transport-remnant powers, and the transport coefficient. The strongest extra-equation diagnostics are dimensionless ratios such as CMB shell width divided by shell center and early-expansion timing ratios. Rounded fit statistics should remain secondary.

The correct development rule is:

$$\text{observe and ablate first; constrain only in the final model revision.} \quad (41)$$

This rule permits denominator-50 structure to remain scientifically useful without becoming a self-fulfilling scoring target.

References

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